A REPORT ON A FMCG RETAIL BUSINESS FROM 2024-2026

BY ASAKOME OGHENAKHOGIE CELESTINE

DATA ANALYST

BRIEF OVERVIEW

- The dataset contains approximately 1800 transaction records.
  - Each row represents a customer purchase transaction.
  - The dataset includes 12 columns covering customer information, product category, price, quantity, and payment method.
  - The dataset simulates real-world retail transaction data.

PROJECT OBJECTIVES

- The main goals of this analysis are:
  - Clean and standardize the dataset
  - Handle missing and invalid values appropriately
  - Create new analytical features such as revenue
  - Analyze sales trends and product category performance
  - Generate insights to support business decision-making

DATA QUALITY ISSUES IDENTIFIED

- Several common data quality problems existed in the dataset:
  - Missing values in age, email, unit_price, quantity, and transaction_date
  - Inconsistent country naming (e.g., Nigeria vs nigeria, USA vs United States)
  - Mixed data types in the unit_price column
  - Negative or invalid quantity values
  - Case inconsistencies in product_category and payment_method

DATA CLEANING METHODS

- All inconsistent namings were corrected and adjusted to a single standard.
- Mixed data types in unit_price column were converted into numbers.
- Negative or invalid quantity values were adjusted.
- Missing values in age and unit _price were handled by using averages while data with missing transaction dates were removed entirely.
- Missing emails were left unattended to.

EXPLORATORY DATA ANALYSIS

- After cleaning the dataset, exploratory analysis focuses on:
  - Sales distribution across countries
  - Most popular product categories
  - Customer age distribution
  - Sales trends over time



FMCG RETAIL BUSINESS

transaction_date
2024 - 2026 YEARS
2024 2025 2026

Total Price $ 4,353,444.35

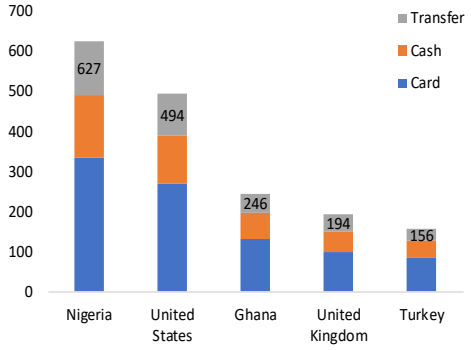
country
- Ghana
- Nigeria
- Turkey
- United Kingdom
- United States

payment_met...
- Card
- Cash
- Transfer

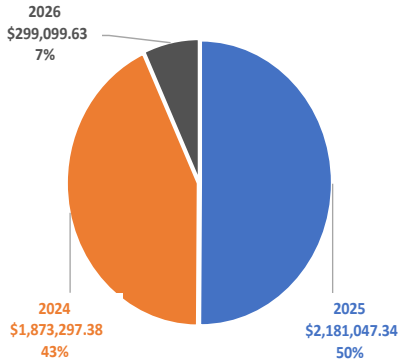
generations
- Baby boomers
- Gen x
- Gen z
- Millennials

product_categ...
- Electronics
- Fashion
- Groceries
- Home

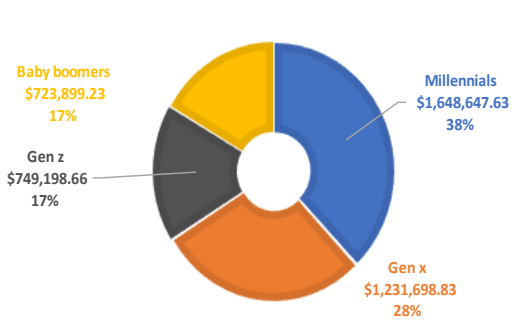
COUNTRY PAYMENT METHOD



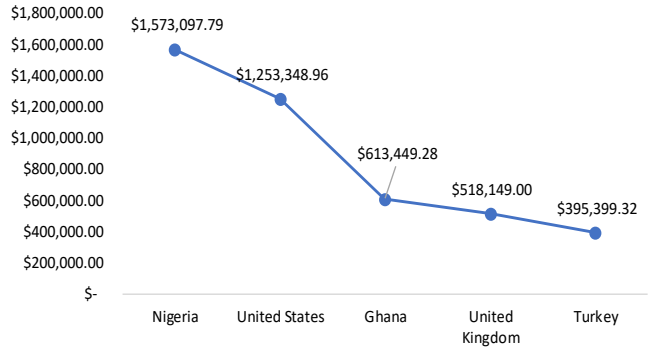
YEARLY TOTAL PRICES



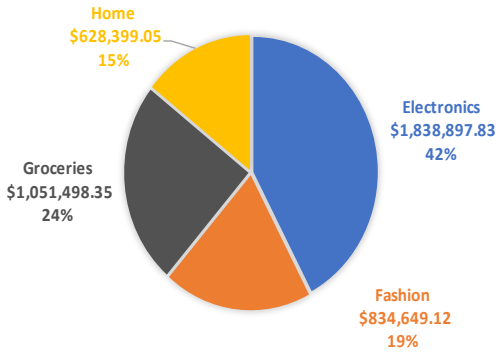
GENERATION TOTAL PRICES



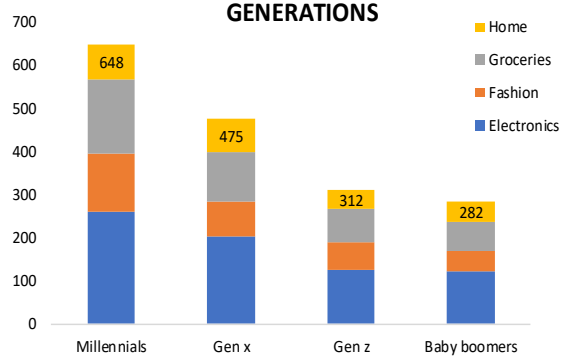
COUNTRY TOTAL PRICES



PRODUCT CATEGORY TOTAL PRICES



PRODUCT CATEGORY COUNT BY GENERATIONS



KEY INSIGHTS AND RECOMMENDATIONS

INSIGHTS

- Nigeria and United States generated the most revenue while Turkey and United Kingdom generated the least revenue
- Electronics were the most sold product category.
- Millennials represent the core customer base.
- Revenue increased yearly.
- Most customers prefer using card as a means for payments.

RECOMMENDATIONS

- Expand the market in Nigeria and United States to maximize profits and investigated why the low revenues in Turkey and United Kingdom.
- Increase inventory for electronics and promote less sold product categories more.
- Target the millennials base and develop strategies to increase other generations engagements.
- Continue using working strategies.
- Increase card payment systems.